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Paleomagnetism of Cretaceous Rocks From Eastern Qiangtang Terrane of Tibet

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Thermal demagnetization has successfully isolated high unblocking temperature characteristic directions from paleomagnetic samples of Lower and Middle Cretaceous redbed formations from eastern Qiangtang Terrane, Tibet. The tilt-corrected mean direction for the Lower Cretaceous is D/I = 57.7°/50.2° with α_{95} = 10.9° and N = 12 sites, and that for the Middle Cretaceous is D/I = 39.7°/51.6° with α_{95} = 8.5° and N = 11 sites. Both formations pass the fold test, and the latter possesses two polarities. The magnetization predates the Eocene and was probably acquired during Cretaceous time. Comparison of the mean directions with expected directions from the stable interior of Eurasia indicates significant clockwise rotation but little paleolatitudinal change for the sampling area. Deformation and rotation of these rocks are presumably related to the collision of India with Asia and its subsequent northward indentation into Asia. Existing paleomagnetic data from Tibet and its environs seems to be compatible with a crustal shortening/thickening model (Dewey et al., 1988) in that the continental crust directly north of the Indian indenter was substantially shortened north-south, whereas in the vicinity of the Western and Eastern Syntaxes, latitudinal displacement was minimal but tectonic rotation was very important. More data are needed to further differentiate between this model and the lateral eastward crustal extrusion model (Molnar and Tapponier, 1975).

INTRODUCTION

The Tibetan Plateau with an average elevation of 5000 m is one of the most prominent topographic features on Earth. This feature, which resulted from the collision of the Indian plate with Eurasia, has stimulated research by Earth scientists from all over the world. Much understanding of the geologic structure and evolution of the plateau has been gained in the last 20 years through the efforts of Chinese scientists as well as joint efforts with their western colleagues, such as the 1980–1982 French-Chinese geoscience expedition to Tibet and the 1985 Royal Society–Academia Sinica Tibet geotraverse. It is now generally accepted that the Tibetan Plateau is an amalgamation of several terranes (Figure 1), most of which probably rifted from Gondwanaland and collided with the Eurasian continent at progressively younger times during the Mesozoic and Cenozoic [*Chang and Cheng*, 1973; *Allegre et al.*, 1984; *Chang et al.*, 1986; *Dewey et al.*, 1988; *Chang et al.*, 1989]. For instance, the Qiangtang Terrane accreted to the Songpan-Garze Terrane, then the southern margin of Eurasia, along the Jinsha Suture during the Late Triassic or Early Jurassic, followed by a Late Jurassic collision of the Lhasa Terrane with the Qiangtang Terrane along the Bangong-Nujiang Suture. These terranes were in turn collided by the India subcontinent along the Indus-Zangbo Suture during the middle Eocene.

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Paleomagnetism is the most effective means of studying terrane motions in the absence of intervening seafloor. For this reason, paleomagnetic studies have been included as an indispensable discipline in all the past major geologic expeditions conducted on the Tibetan Plateau. Indeed, paleomagnetic investigations of the Lhasa Terrane have revealed convergence of as much as 1900 ± 850 km between this terrane and Eurasia since Late Cretaceous time and differential, counterclockwise rotations of up to 41.5° ± 12.5° with respect to Eurasia within the terrane [*Achache et al.*, 1984]. It is also of interest to note that *Lin and Watts* [1988] determined a 22° south paleolatitude for the Kunlun (Songpan-Garze) Terrane for Carboniferous time.

Paleomagnetic investigations conducted so far are confined to the western part of the plateau [*Achache et al.*, 1984; *Lin and Watts*, 1988]. To better understand the tectonic evolution of this vast area, more studies from different regions are needed. The present paper reports paleomagnetic results from Cretaceous rocks from the eastern part of the Qiangtang Terrane.

GEOLOGIC SETTING AND SAMPLING

The Qiangtang Terrane is separated from the Songpan-Garze Terrane to the north by the Jinsha Suture and from the Lhasa Terrane to the south by the Bangong-Nujiang Suture (Figure 1). West of the 96°–98°E meridian, the Jinsha and Bangong-Nujiang sutures trend west to west-northwest; east of 96°–98°E, both sutures sharply turn to a south-southeastern direction. Accordingly, the western and central

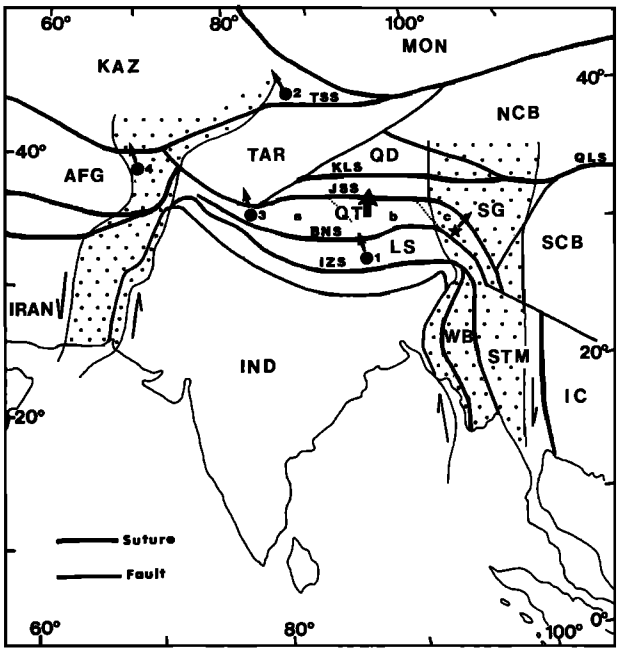


Fig. 1. Schematic tectonic map of Tibet and its environs (modified from *Dewey et al.* [1988]). Major terranes: AFG, Afghanistan; IC, Indochina; IND, India; KAZ, Kazakhstan; LS, Lhasa; NCB, North China Block; MON, Mongolia; QD, Qaidam; QT, Qiangtang; SCB, South China Block; SG, Songpan-Garze (Complex); STM, Shan-Thai-Malay; TAR, Tarim; WB, West Burma. Major sutures: BNS, Bangong-Nujiang; IZS, Indus-Zangbo; JSS, Jinsha; KLS, Kunlun; QLS, Qinling; TSS, Tienshan. The numbered solid circles denote areas where Cretaceous rocks were sampled and studied by other paleomagnetists: 1, *Achache et al.* [1984]; 2, *Chen et al.* [1991*a*]; 3, *Chen et al.* [1991*b*]; and 4, *Pozzi and Feinberg* [1991]. The sampling area of this study is marked with a star. Observed mean declination at each sampling locality is indicated by a small arrow. Large arrow indicates expected direction of declination in Tibet for the Cretaceous (110 Ma). The dotted lines within the Qiangtang Terrane mark the rough boundaries of the subdivisions of the Terrane; a, the Qiantang Block (sensu stricto); b, the Tanggula-Kaixinling Depression; and c, the Qamdo Block. The stippled areas indicate the shear zones that bound the Indian indenter [*Dewey et al.*, 1988].

parts of the Terrane are east-west striking, whereas the eastern part of it bends into north-northwestern orientation. Farther south, it becomes narrow in width and finally joins with the Yunnan (Eastern) Syntaxis [*Tapponier and Molnar*, 1977; *Sengor*, 1984; *Wang and Chu*, 1988]. Whether the Shan-Thai or Indochina terrane represents the southern extension of the Qiangtang Terrane across the Syntaxis is controversial [*Mitchell*, 1989; *Chang et al.*, 1989].

On the basis of stratigraphic data and structural features, *Chang et al.* [1989] recently proposed that the Qiangtang Terrane can be subdivided, from west to east, into the Qiangtang Block sensu stricto, Tanggula-Kaixinling Depression, and Qamdo Block (Figure 1). The Qiangtang Block sensu stricto has a metamorphic-crystalline basement of probably Precambrian age, on which platform-type Paleozoic sediments have been deposited since the Silurian. The presence of Gondwanan facies have been confirmed in the Upper Carboniferous to Lower Permian in southeastern Karakorum. Cathaysian flora and Tethyan fauna appear to be dominant in the Upper Permian sequences. The Mesozoic record is characterized by shallow marine sediments, the Paleogene record is characterized by thick molassic deposits, and the Neogene record is characterized by gypsum-bearing red beds. In the Tanggula-Kaixinling Depression, a thick accumulation of Late Paleozoic to Tertiary sediments and volcanics is present. Tertiary red beds, which overlie the Jurassic rocks, were involved in strong folding and northward thrusting related to the collision of India with Asia. The Qamdo Block is bounded on the east by the Jinsha Suture and on the west by the Taniantaweng orogenic arc. The oldest exposed strata are Lower Ordovician in age. A pronounced unconformity separates the Lower Ordovician rocks from a Middle Devonian to Upper Permian fossil-bearing sequence of littoral and shallow marine facies. The faunas have a strong Tethyan affinity and the late Permian flora contains typical *Cathaysiopteris*, implying a relatively close relation between the Qamdo Block and the South China (Yangtze) Block during the late Paleozoic. Lower and Middle Triassic sequences are littoral-shallow-marine clastic and carbonate rocks with intercalated intermediate-acidic volcanic rocks. The Upper Triassic rocks include, from bottom to top, nonmarine red clastic rocks, shallow marine carbonate rocks, and nonmarine clastic rocks with coal seams. Comformably overlying the Triassic sequences is a thick red bed sequence of Jurassic to Cretaceous age, which is in turn disconformably overlain in places by Eocene-Oligocene red molassic deposits or Pliocene terrestrial trachytes and clastic rocks.

Like the red beds in western Yunnan province, the Qamdo Juro-Cretaceous red bed sequence is well known for its great thickness and wide areal distribution in the Qamdo-Markam area [*Ren et al.*, 1980]. The whole sequence is exposed in a synclinorium stretching from Qamdo south-southeastward to Markam. The sequence was previously loosely divided into the Middle-Lower Jurassic Chaya Group and the Upper Jurassic to Lower Cretaceous Xiangdui Group. In the last few years, geologists from the No. 3 Geologic Brigade of Yunnan province who were mapping in the Markam area on a scale of 1:200000 found that the sequence is correlatable with the West Yunnan red beds in terms of lithology and paleontology. They redivided the sequence into Jurassic and Cretaceous systems, the latter of which was further divided into the Lower Cretaceous Cuowa Formation (Berriasian-Barremian), the Middle Cretaceous Mangkang (Markam) Formation (Aptian-Turonian), and the Upper Cretaceous Xuzhong Formation (Coniacian-Maastrichtian) in accord with a three-fold subdivision scheme for the Chinese Cretaceous System proposed by *Hao et al.* [1986].

In the vicinity of Markam county the Lower Cretaceous Cuowa Formation is more than 1900 m thick and conformably rests on the Upper Jurassic Rumei Formation. It consists of light grey sandy conglomerate and arkosic quartz sandstone in the basal part and upward chiefly of purplish-red fine-grained sandstone and siltstone with intercalations of silty mudstone and calcareous conglomerate. Some bivalves such as *Peregrinoconcha yunnanensis* Chen et Lan, *Koreanaia antigua* (Gu et Ma), and *Nippononaia diana* Guo have been found in the Cuowa Formation. Vertebrate fossils *Microvenator chagyabi* Chao, *Monkonosaurus lawulacus* Chao, and *Prodeinodon tibetensis* Chao were reported earlier from identical horizons [see *Hao et al.*, 1986]. The overlying Mangkang Formation is more than 1000 m thick and is composed of brick-red sandstone interbedded with siltstone and conglomerate. It yields ostracods *Eucypris profunda* (Lubimova), *E. angulata* Ye, *Darwinula contracta*

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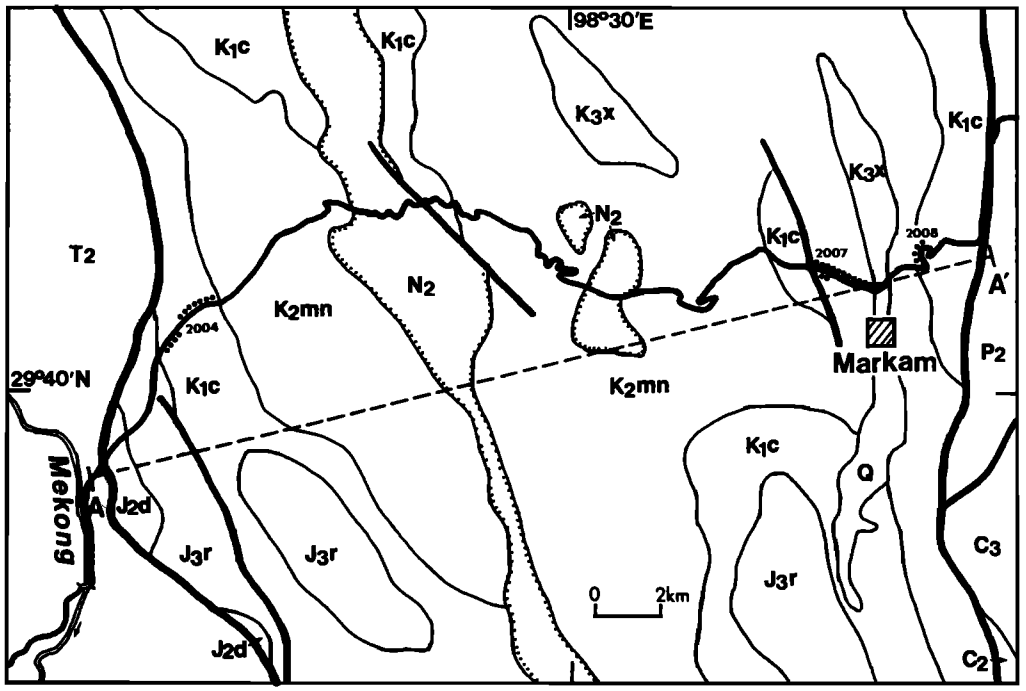


Fig. 2. Geologic map of the Markam area. Medium line is the road from Chengdu to Lhasa, double line is the Mekong (Lancang) River, and wide lines are faults. Numbered dots denote site locations. A-A′ is the location of the cross section shown in Figure 3. Formation names: Q, Quaternary; N_2, Pliocene; K_3x, Upper Cretaceous Xuzhong Formation; K_2mn, Middle Cretaceous Mangkang Formation; K_1c, Lower Cretaceous Cuowa Formation; J_3r, Upper Jurassic Rumei Formation; J_2d, Middle Jurassic Duiba Formation; T_2, Middle Triassic; P_2, Upper Permian; C_3, Upper Carboniferous; C_2, Middle Carboniferous.

Mandelstam, *Cypridea sp.*, *Quadrocypridea regularis sp. nov.*, and *Cristocypridea demiorbiculata* Ye and is thought to be correlative with the Nanxing or Mangang formations in western Yunnan province.

Sampling was carried out at three localities near the town of Markam along the southern route of the road from Chengdu to Lhasa (Figure 2). At the first locality 20 km west of Markam, 11 sites (2004A–K) were drilled in the Cuowa Formation on the west limb of the synclinorium (Figure 3), where the bedding attitude varies from almost vertical to overturned with dips at 80°–125° to the east. Four sites (2008A–D) were drilled at the second locality 3 km northeast of Markam in the same formation on the east limb of the synclinorium, where the bedding is overturned and dips at 107°–115° to the west. Thirteen sites (2007A–M) were drilled in the Mangkang Formation at a third locality 3 km west of the second one. These sites were distributed on two limbs of an anticline within the synclinorium, which dips 45°–60° toward the east and 44°–60° toward the west, respectively. Therefore it is possible to perform a fold test on both formations.

Five cores were usually taken from each site representing one horizon. A portable rock drill was used for sample collection, and an orientation device mounted with a magnetic compass was used for core orientation.

TREATMENT AND ANALYSIS

Standard magnetic measurement and demagnetization procedures were conducted at the University of Florida. Since all the samples are red beds, thermal demagnetization was used throughout the investigation. Stepwise demagnetization was performed in a TSD-1 Schonstedt thermal demagnetizer, and remanence of the sample was measured in an ScT two-axis cryogenic magnetometer. All the equipment is housed in a magnetically shielded room with a residual field of 50–250 nT. The initial pilot demagnetization study employed temperature increments of 100° below 400°C, 50°

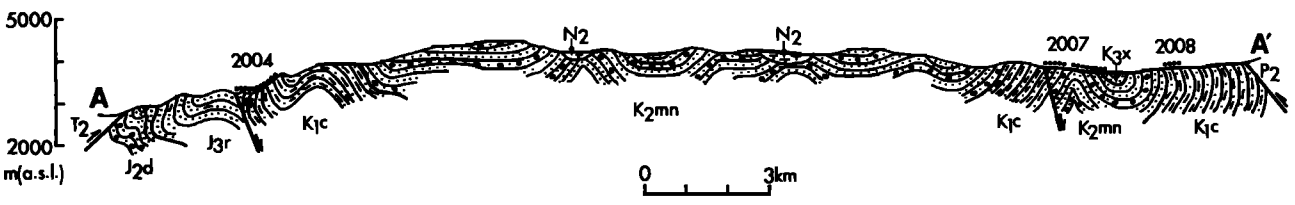


Fig. 3. A-A′ cross section showing the geologic structure and its relation to the sampling sites. The vertical scale indicates elevation in meters above sea level. Formation names as in Figure 2.

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TABLE 1. Site and Formation Statistics for the Cuowa Formation

Site	Bedding	Component	Temperature, °C	n/N	In Situ D, deg	In Situ I, deg	Tilt Corrected D, deg	Tilt Corrected I, deg	α_{95}	k_2	Pole Position Latitude, °N	Pole Position Longitude, °E
				Locality 2004 (29.7°N, 98.4°E)								
A	350/80	HT	680–710	4/5	268.6	46.6	70.2	52.7	9.0	104.5	31.2	165.3
		LT	0–600	4/5	0.8	30.8.	22.3	−4.0	24.0	15.6		
B	0/84	HT	620–705	5/5	279.2	42.1	78.6	53.0	9.0	73.2	24.7	162.4
		LT	0–600	4/5	347.3	54.5	54.6	12.3	14.9	14.9		
C*	335/120	HT	600–700	5/5	292.9	19.7	16.0	22.3	20.7	14.6	66.6	235.6
		LT	0–600	5/5	355.1	40.9	6.5	−33.5	16.2	23.4		
D	335/120	HT	640–705	5/5	259.7	5.6	40.9	51.7	11.4	46.2	55.0	173.3
		LT	0–600	5/5	342.8	34.7	3.1	−22.4	23.0	12.0		
E	336/125	HT	620–700	5/5	265.4	5.8	38.1	45.3	5.4	198.9	56.5	183.9
		LT	0–600	4/5	19.9	49.7	14.1	−53.6	47.8	4.7		
F*	340/124	HT	620–710	4/5	267.7	16.0	48.5	37.3	29.7	10.5	45.6	188.3
		LT	0–685	5/5	349.1	54.1	26.9	−32.1	47.5	3.6		
G	330/120	HT	670–705	5/5	280.6	16.5	14.4	29.3	13.7	32.2	70.7	231.9
		LT	0–600	5/5	350.0	45.1	6.5	−34.3	24.0	11.1		
H*	342/120	HT	640–690	3/5	255.9	14.5	66.6	45.3	37.4	12.0	32.1	173.6
		LT	0–600	3/5	340.8	42.5	20.9	−18.9	22.3	31.7		
I	324/120	UN	400–685	5/5	276.1	23.7	46.0	31.6	7.9	95.1	46.2	194.5
J	351/83	HT	600–705	5/5	248.2	47.2	94.1	48.3	12.1	41.0	10.9	160.7
		LT	0–400	5/5	1.9	50.3	42.2	−1.6	11.1	48.5		
K	350/85	HT	670–700	5/5	250.9	55.0	86.8	39.5	14.4	29.1	13.5	170.1
		LT	0–600	5/5	7.2	51.6	43.3	−6.6	28.0	8.4		
				Locality 2008 (29.7°N, 98.6°E)								
A	208/115	UN	400–695	3/5	64.3	−51.6	75.9	41.7	2.6	2272.3	23.3	173.3
B	200/115	UN	400–705	4/5	76.2	−50.1	72.7	53.8	7.8	138.3	29.5	163.6
C	186/107	UN	400–690	5/5	70.6	−33.9	47.4	61.7	6.2	152.7	50.2	156.1
D	190/113	UN	400–690	5/5	67.9	−29.4	29.4	60.6	3.3	539.6	63.5	153.9
				Formation Mean								
		HT+UN†		12/15	280.9	18.4	57.7	50.2	10.9		40.6	170.5
		LT‡		10/10	355.1	46.0	25.1	−20.3	15.9			

Bedding, strike/dip with dip direction 90° clockwise from strike; HT, LT, UN, high-temperature, low-temperature, and univectorial component, respectively; n/N, number of samples or sites in calculation for the means/number of samples or sites measured; D, I, declination and inclination; α_{95}, A_{95}, radius of the circle of 95% confidence about the mean direction and pole, respectively; k_1, k_2, Fisher precision parameters before and after tilt correction.

*Excluded from calculation of the formation mean.

†k_1 = 1.4, k_2 = 16.7, and A_{95} = 13.0.

‡k_1 = 48.7 and k_2 = 10.2.

between 400° and 550°C, 25° between 550° and 600°C, 10° in the range 600°–680°C, and finally 5° increments to the destruction of sample remanence. Abbreviated steps were applied to the remaining samples. At temperatures above 670°C, time of heating appeared to be crucial in demagnetizing the high-temperature component of magnetization of these samples. Often, 45–50 min were needed.

Orthogonal vector plots were constructed for each specimen to examine its demagnetization behavior [*Zijderveld*, 1967]. Principal component analysis [*Kirschvink*, 1980] was employed to compute paleomagnetic directions and planes. For characteristic remanent magnetization (ChRM), the directions were calculated both with and without anchors to the origin. In most cases, the associated maximum angular deviation (MAD) values were small, and the directions calculated in both ways were similar, indicating that the ChRM directions were well defined. Site and formation means were computed using *Fisher* [1953] statistics. Sector-constrained great circle intersection [*McFadden and McElhinny*, 1988] was used for one site. Sample directions with MAD values or site mean directions with α_{95} values greater than 15° were discarded for the calculation of site or formation means. Only means based on directions not anchored to the origin are listed in Tables 1 and 2.

PALEOMAGNETIC RESULTS

Lower Cretaceous Cuowa Formation (K_1c)

The intensities of natural remanent magnetization (NRM) of the samples from this formation varied from 0.7 to 13.4 mA/m with the majority in the range 1.0–3.0 mA/m. Samples from two sites (2008C and 2008D) were more strongly magnetized with NRM intensities ranging from 4.1 to 13.4 mA/m. Progressive thermal demagnetization revealed two components of magnetization at all sites except site 2004I from the west limb (Figures 4*a*–4*d*). A low to middle temperature component (LTC), which accounted for 10–60% of the initial intensity, was often destroyed below 600°C. Although scattered, the in situ LTC directions were similar to that of the present-day field, with a mean of D = 355.1°, I = 46.0°, N = 10 (sites) and k_2/k_1 = 0.21 (see Figure 5 and Table 1), a failed fold test indicative of a recent post-folding origin. Since the LTC persisted up to 640°C in some

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TABLE 2. Site and Formation Statistics for the Mangkang Formation

Site	Bedding	Component	Temperature, °C	n/N	In Situ D, deg	In Situ I, deg	Tilt Corrected D, deg	Tilt Corrected I, deg	α_{95}	k_2	Pole Position Latitude, °N	Pole Position Longitude, °E
Locality 2007 (29.7°N, 98.7°E)												
A*	185/50	UN	600–695	5/5	39.6	16.8	14.2	37.1	27.1	9.0	74.3	220.4
B	174/60	UN	450–690	5/5	45.7	−9.3	34.8	36.2	5.0	232.7	57.2	197.5
C	130/60	UN	450–700	5/5	21.5	2.6	3.9	57.5	8.0	93.6	81.0	118.8
D	353/62	UN	450–690	5/5	298.9	44.2	32.3	57.2	3.9	389.2	62.1	163.1
E	30/45	UN	400–690	5/5	9.8	38.7	49.0	39.3	3.7	433.6	45.7	186.3
F	30/47	UN	450–690	5/5	337.4	45.2	47.8	63.3	10.7	52.4	49.7	153.0
G	56/46	UN	600–700	5/5	5.5	28.7	43.4	55.1	9.5	66.3	53.3	167.7
H	40/50	UN	600–690	5/5	10.8	36.8	56.6	43.3	8.1	91.3	40.1	179.4
I*	10/52	HT	670–700	5/5	230.9	−67.9	261.9	−21.9	19.5	16.3	−12.6	2.6
				5/5†	151.2	−60.5	242.3	−51.0	3.2	1233.6	−37.2	349.5
		LT	0–680	5/5	308.8	46.3	40.6	67.3	4.2	332.9		
J	2/60	UN	600–690	5/5	325.2	49.7	44.1	45.8	3.4	513.4	51.4	181.1
K	2/60	UN	600–690	9/10	323.2	52.1	47.6	46.7	3.9	178.3	48.5	178.8
L*	168/44	UN	600–700	5/5	60.9	15.2	47.5	56.0	15.6	24.9	50.0	165.9
M	176/49	UN	450–690	5/5	44.6	32.1	357.6	55.9	7.6	102.3	82.9	82.6
Formation Mean‡												
				11/13	1.3	39.9	39.7	51.6	8.5		56.7	172.7

See Table 1 for other explanations.
*Excluded from calculation of the formation mean.
†Determined using *McFadden and McElhinny*'s [1988] method.
‡k_1 = 5.8, k_2 = 29.8, and A_{95} = 10.6.

cases, fine-grained secondary hematite is the most probable carrier of the LTC. A high-temperature characteristic component (HTC), which has a northeasterly and down direction in stratigraphic coordinates, was resolved above 620°C and often unblocked between 660° and 700°C. In some cases, unblocking extended even up to apparent temperatures of 710°C (see Figures 4*a*–4*d*). Samples from one site from the west limb and the four sites from the east limb of the synclinorium showed basically univectorial decay during thermal demagnetization up to 695°C, although in some cases a small, presumably viscous magnetization was removed below 200°C (Figures 4*e* and 4*f*). Hematite is the principal magnetic mineral responsible for both the HTC and univectorial components. Site mean directions from the two limbs calculated from the HTC and univectorial components are almost antipodal before tilt correction, with the west limb group having westerly and down directions and the east limb group having east-northeasterly and up directions. After correcting for bedding tilt, however, all the site mean directions converge upon a mean in the northeast quadrant (Figure 6), resulting in an increase in k value from 1.4 to 16.7. Hence the ChRM directions pass the fold test at the 99% confidence level [*McElhinny*, 1964], which suggests a prefolding age for the ChRM. The tilt-corrected mean direction for the Cuowa Formation is D = 57.7°, I = 50.2°, and α_{95} = 10.9° (Table 1).

Middle Cretaceous Mangkang Formation (K_2mn)

Samples from the Mangkang Formation were generally more strongly magnetized than those from the Cuowa Formation, with the majority of initial NRM intensities in the range of 2.0–7.0 mA/m. The demagnetization behavior of most samples from this formation was similar to that of the univectorial group of the Cuowa Formation. At temperatures above 200°C, magnetizations of most samples monotonously decayed toward the origin to temperatures up to 690°C (Figures 7*a*, 7*b*, and 7*d*). The in situ directions of these univectorial components deviate from the present-day geomagnetic field direction at the sampling locality. Most of them are of normal polarity, with northeast declinations and positive inclinations after bedding correction. A reversed component with a southwesterly and up direction in stratigraphic coordinates was consistently resolved from samples from one site (2007I) in the temperature range of 680°–700°C (Figure 7*c*). The reversed direction calculated using principal component analysis technique varies from sample to sample, resulting in a shallow inclination and a large α_{95} value for the mean direction of this site. On the other hand, samples from site 2007I exhibited good remagnetization circle paths in stereographic projections. Application of sector-constrained great circle intersection [*McFadden and McElhinny*, 1988] retrieved a site mean direction which has a much smaller α_{95} value and is nearly antipodal to the normal polarity group (Figure 8 and Table 2). Similar cases have been seen in Cretaceous red beds from western Yunnan province (our own unpublished data) and Upper Jurassic rocks from southwestern Sichuan province [*Huang and Opdyke*, 1991]. Therefore the mean direction of this site determined using McFadden and McElhinny's method was used in the calculation of the formation mean. A fold test that is positive at the 99% confidence level (k_2/k_1 = 5.15, N = 11) [*McElhinny*, 1964] indicates that the ChRM is prefolding (Figure 9). The tilt-corrected mean direction for the Mangkang Formation is D = 39.7°, I = 51.6°, α_{95} = 8.5° (Table 2).

DISCUSSION

Thermal demagnetization has revealed a postfolding low unblocking temperature component and a prefolding characteristic component at high temperatures from the Cuowa Formation. Similarly, the univectorial ChRM isolated from

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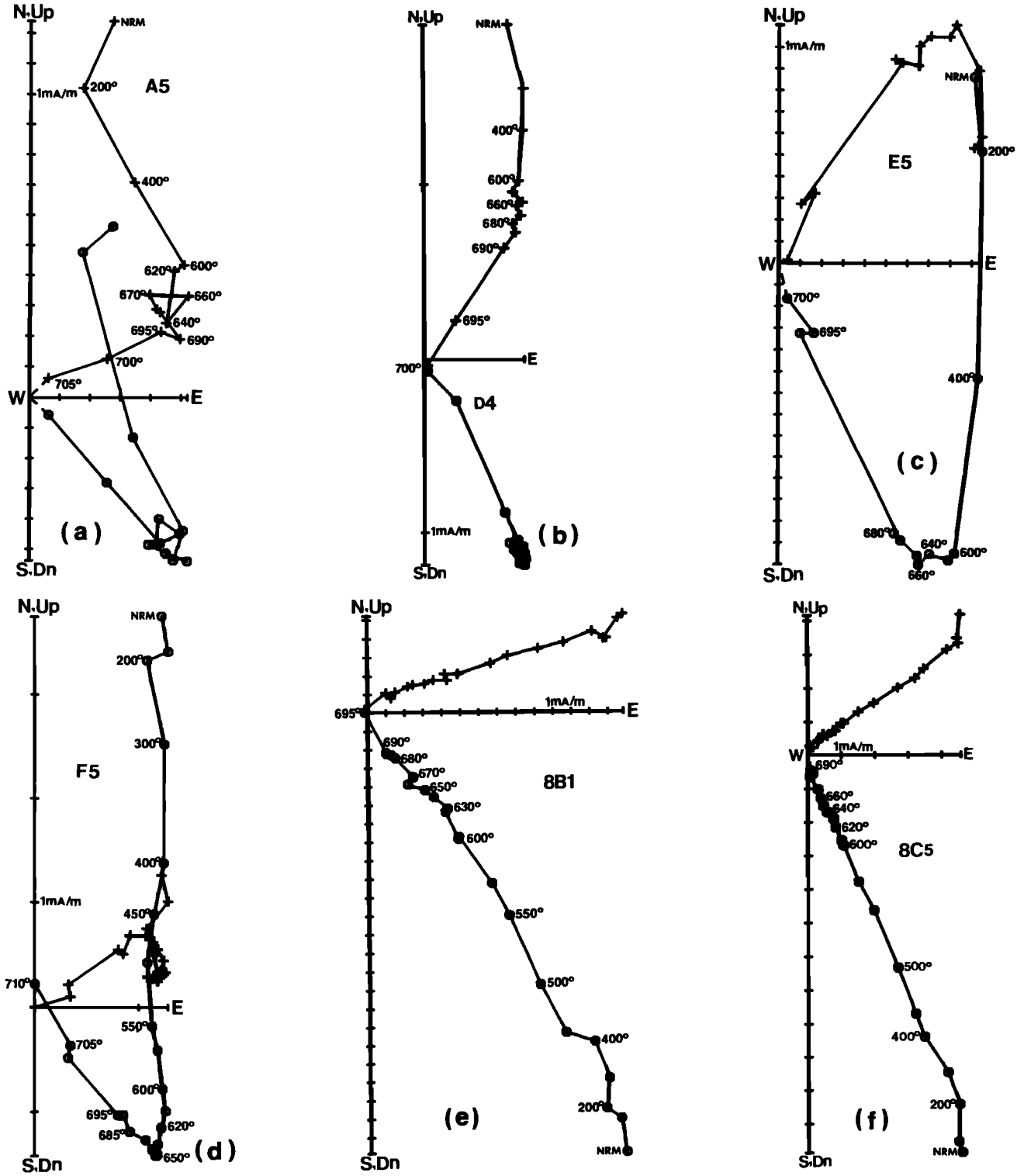


Fig. 4. Representative orthogonal plots for the samples from the Lower Cretaceous Cuowa Formation. Samples A5, D4, E5, and F5 are from the western limb of a synclinorium, and samples 8B1 and 8C5 are from the eastern limb. All directions are tilt-corrected. Crosses (circles) are plotted on horizontal (vertical) planes. Thermal treatment levels are in degrees Celsius.

the Mangkang Formation has been demonstrated to be prefolding. The folded Mesozoic rocks of the study area are overlain disconformably by Eocene-Oligocene molassic deposits. Therefore the ChRMs in both formations can be dated as pre-Eocene. Since India collided with Asia during Eocene time [*Allegre et al.*, 1984; *Dewey et al.*, 1988], the acquisition of the ChRMs of both formations predates the collision.

Our results are substantiated by the work of others. *Otofuji et al.* [1990] reported a ChRM direction from five

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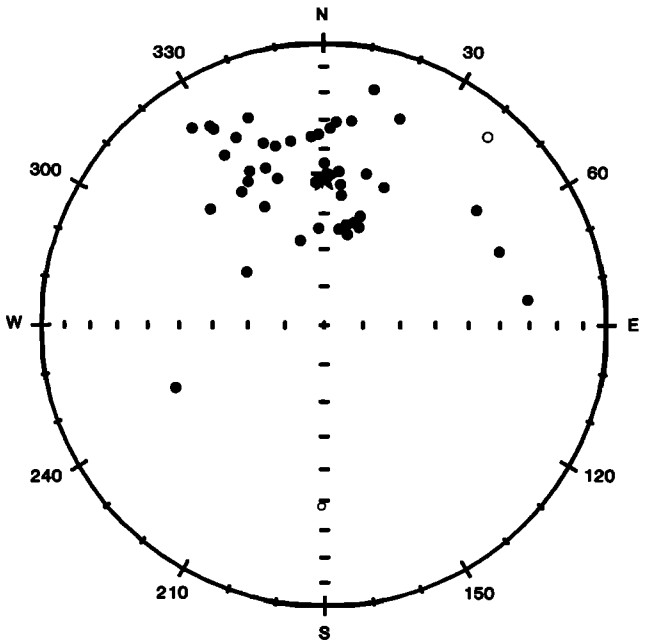


Fig. 5. Equal-area projections of the in situ low-temperature component directions of the samples from the Cuowa Formation. Solid (open) symbols represent directions plotted onto the lower (upper) hemisphere. The star denotes the present-day geomagnetic field direction at the sampling locality.

sites at the same sedimentary section. The direction they observed was $D = 48.2°$, $I = 49.0°$, $\alpha_{95} = 8.8°$, in agreement with ours.

Acquisition of magnetization of red beds often involves more than one process and may occur over a long period of time [*Cox and Hart*, 1986], so it is difficult to place an accurate date for the ChRMs isolated from the two formations. The Mangkang Formation was mapped as Middle Cretaceous (Aptian to Turonian) in age, and the deposition of the formation would be entirely within the Cretaceous Long Normal Superchron [*Harland et al.*, 1989]. However, one of the 13 sites from this formation is reversely magnetized. It is most probable that the age of the ChRMs of the two formations is Cretaceous but that the ChRM of the Cuowa Formation might be as late as the Cretaceous Long Normal Superchron and that of the Mangkang Formation might be late as the Late Cretaceous.

We now compare our observed directions with expected directions from stable Eurasia. We choose the synthetic apparent polar wander path (APWP) for Eurasia proposed by *Besse and Courtillot* [1991] as a reference. *Besse and Courtillot* [1991] selected paleomagnetic poles with more stringent criteria from the large database now available and used oceanic kinematic models to transfer the selected poles to a common geographic frame. They then averaged the transferred poles in windows of 20 m.y. to generate the synthetic APWP. The observed and expected directions for the sampling locality (29.7°N, 98.7°E) are presented in Table 3 and Figure 10. Note that paleomagnetic rotations were determined by subtracting expected declinations from observed ones and that latitude differences were calculated by subtracting observed paleolatitudes from expected ones. Therefore positive rotations are clockwise, and positive latitude differences indicate northward translation.

These calculations indicate significant clockwise rotation but insignificant latitudinal change of the sampling locality after the Cretaceous. The sense of rotation is compatible with post-Eocene left-lateral slip along the major east trending faults in northeastern Tibet, which has been recognized from satellite imaginery, earthquake fault plane solutions, and indentation experiments with plasticine [*Molnar and Tapponnier*, 1975; *Tapponnier and Molnar*, 1977; *Tapponnier et al.*, 1982]. If rotation was triggered by the collision of India with Asia 45 m.y. ago [*Dewey et al.*, 1988], the rotation

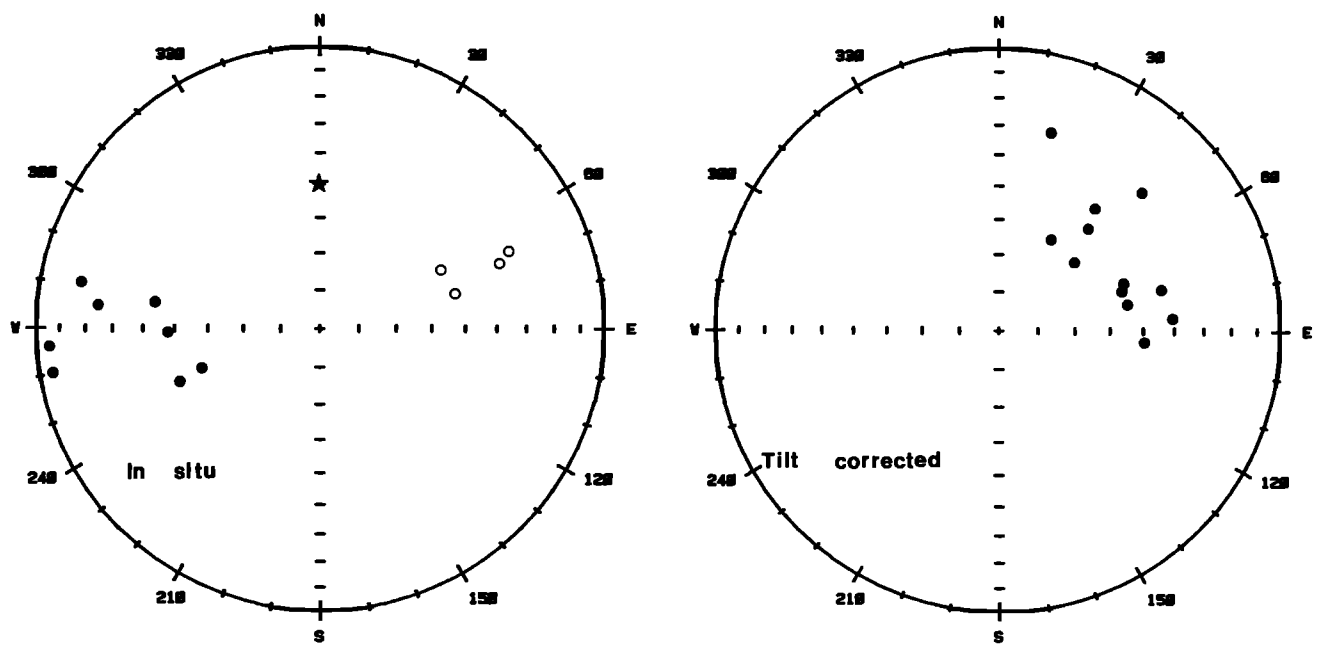


Fig. 6. Equal-area projections of the site mean directions for the Cuowa Formation before and after tilt correction. Symbols as in Figure 5.

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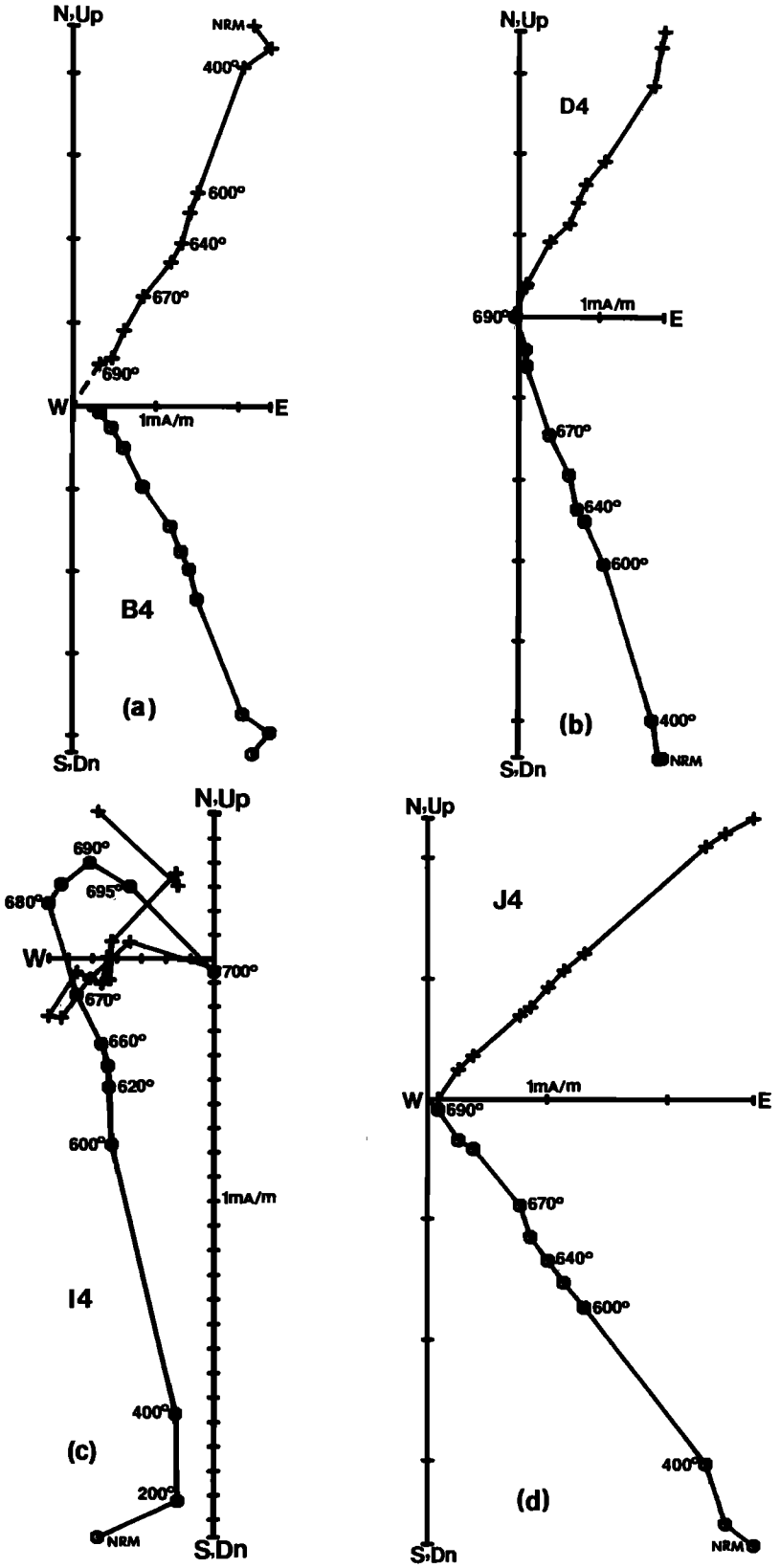


Fig. 7. Representative orthogonal plots for the samples from the Middle Cretaceous Mangkang Formation. Sample B4 is from the western limb of an anticline and others are from the eastern limb. Coordinates and symbols as in Figure 4.

rate would be in the range 0.5°–1.0° m.y.$^{-1}$. The insignificant latitudinal change derived from this data set would imply that insignificant crustal shortening has taken place north of the eastern Qiangtang Terrane since Cretaceous time.

The present thickness of the crust in the Markam area is about 61 km [see *Zhang et al.*, 1984, Figure 10], nearly twice that of normal continental crust. Strong folding and northward thrusting observed in the Paleogene red beds in and

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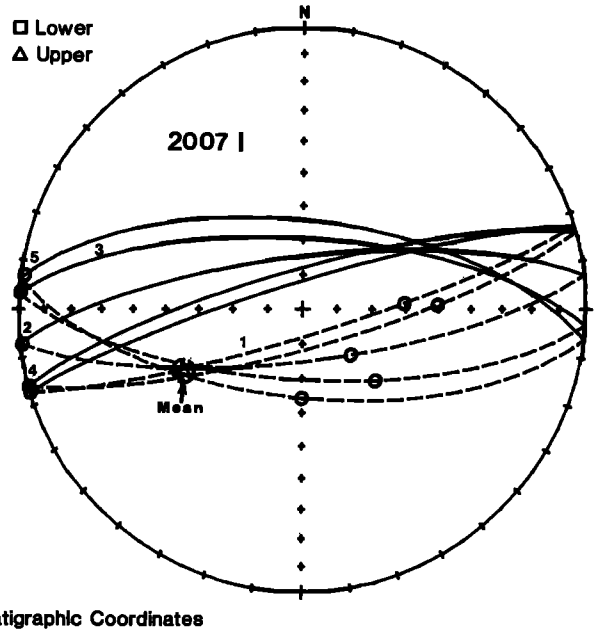


Fig. 8. Great circle intersection for site 2007I. Sector constraints are marked with small open circles, and numbers along the great circle paths are sample numbers.

north of the Qiangtang Terrane along the Royal Society–Academia Sinica geotraverse [*Chang et al.*, 1986] are evidence for post-Eocene north-south crustal shortening in and north of the central part of the terrane. Shortening by folding alone since the Eocene is estimated to be at least 40% [*Chang et al.*, 1989]. Moreover, about 1980 km of intracontinental convergence is estimated to be absorbed north of the Indus-Zangbo Suture [*Achache et al.*, 1984; *Lin and Watts*, 1988; *Dewey et al.*, 1988]. About 390 and 670 km are estimated to be partitioned in the Tienshan north of the Tarim Basin [*Dewey et al.*, 1989], and in the Altay and Sayan Tuva ranges north of the Dzungaria Basin [*Chen et al.*, 1991*a*], respectively. It seems clear from these geologic and paleomagnetic observations that substantial north-south crustal shortening has taken place in the central part of the Tibetan Plateau since the collision of India with Asia. Why then is there no significant latitudinal crustal shortening indicated by the paleomagnetic results from northeastern Tibet north and east of the Eastern Syntaxis as reported here?

Two sets of paleomagnetic data were recently reported from the Lower Cretaceous rocks on the eastern and western sides of the Pamir (Western) Syntaxis [*Chen et al.*, 1991*b*; *Pozzi and Feinberg*, 1991]. A prefolding Cretaceous direction derived from 14 sites along the Xinjiang-Tibet highway yielded a rather low paleolatitude of 10.9° ± 3.0° for a reference at Domar (33.7°N, 80.4°E) [*Chen et al.*, 1991*b*] but indicated little rotation relative to cratonic Eurasia (Y. Chen, personal communication, 1991). Another study was carried out by *Pozzi and Feinberg* [1991] on the Lower Cretaceous red beds from the Tajik basin (38.6°N, 68.9°E). They isolated a stable and well-defined magnetization from most samples of all four sites and a second direction of high unblocking temperatures and reversed polarity from several samples of one site. They interpreted the former as a prefolding remagnetization and the latter as a primary direction. The results suggested a counterclockwise rotation of 26° ± 9° for the sampled area, and insignificant (probably 100–300 km) postcollision crustal shortening north of the Tajik basin.

A general picture outlined by the existing paleomagnetic data appears to be that immediately north of the Indian indenter the continental crust was substantially shortened in a north-south direction. In the western and eastern zones bounding the indenter, however, tectonic rotation, counterclockwise in the western and clockwise in the eastern zone, was prominent and north-south crustal shortening was insig-

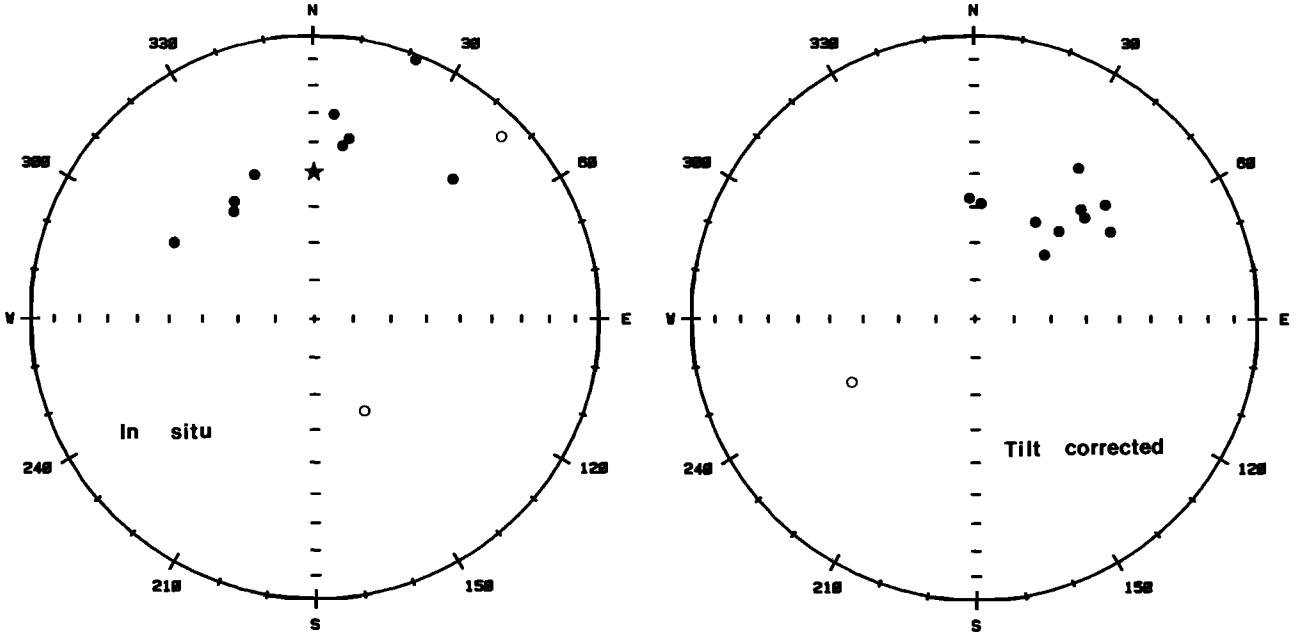


Fig. 9. Equal-area projections of the site mean directions for the Mangkang Formation before and after tilt correction. Symbols as in Figure 5.

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TABLE 3. Calculated Rotation and Latitude Difference of the Sampling Locality With Respect to Stable Eurasia

Age, Ma	D_{exp}, deg	Rotation, deg	P_{exp}, deg	Latitude Difference, deg
140	21.2 ± 12.4	36.5 ± 21.0 (C)	33.0 ± 10.4	2.0 ± 14.3 (C)
130	15.6 ± 5.5	42.1 ± 17.9 (C)	24.4 ± 5.0	−6.6 ± 11.0 (C)
120	15.3 ± 4.5	42.4 ± 17.6 (C)	23.1 ± 4.1	−7.9 ± 10.6 (C)
		24.4 ± 14.4 (M)		−9.1 ± 8.9 (M)
110	17.4 ± 5.6	40.3 ± 17.8 (C)	23.5 ± 5.1	−7.5 ± 11.0 (C)
		22.3 ± 14.8 (M)		−8.7 ± 9.4 (M)
100	14.8 ± 5.6	42.9 ± 17.9 (C)	26.9 ± 5.0	−4.1 ± 11.0 (C)
		24.9 ± 14.8 (M)		−5.3 ± 9.3 (M)
90	14.6 ± 3.6	43.1 ± 17.4 (C)	26.3 ± 3.2	−4.7 ± 10.3 (C)
		25.1 ± 14.2 (M)		−5.9 ± 8.5 (M)
80	15.2 ± 3.8	24.5 ± 14.2 (M)	26.4 ± 3.4	−5.8 ± 8.6 (M)
70	14.5 ± 4.6	25.2 ± 14.5 (M)	28.1 ± 4.1	−4.4 ± 8.9 (M)

Cuowa Formation, D_{obs} = 57.7° ± 17.0°, P_{obs} = 31.0° ± 9.8°; and Mangkang Formation, D_{obs} = 39.7° ± 13.7°, P_{obs} = 32.2° ± 7.9°. D_{obs}, P_{obs}, observed declination and paleolatitude. D_{exp}, P_{exp}, expected declination and paleolatitude from Eurasian APWP of *Besse and Courtillot* [1991]. Rotation and latitude difference calculated from the Cuowa and Mangkang formations are specified with (C) and (M), respectively. Estimation of uncertainties follows *Beck* [1980] and *Achache et al.* [1984].

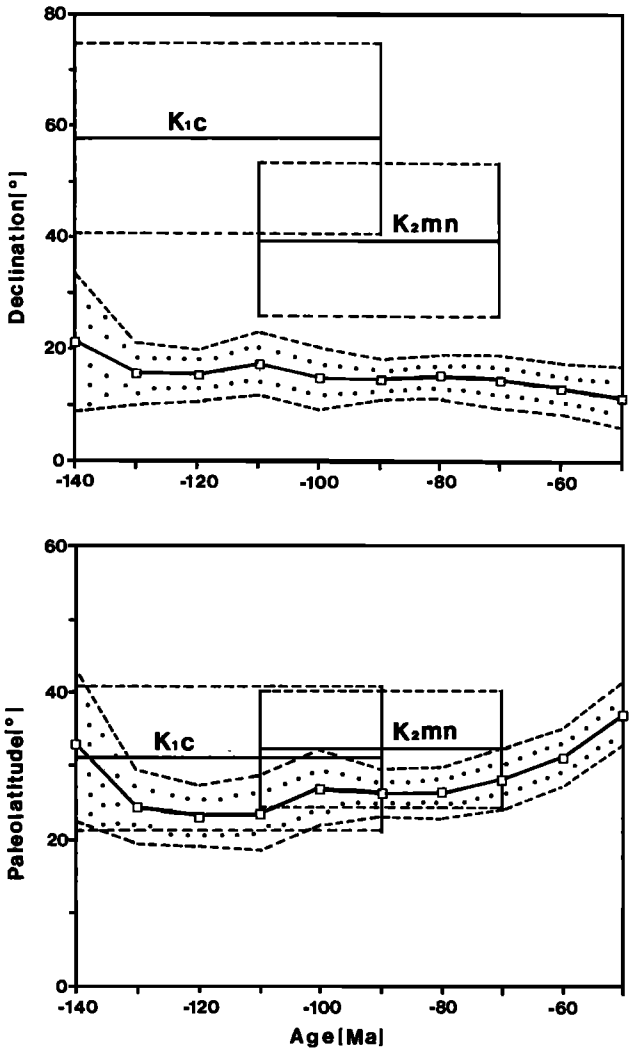


Fig. 10. Predicted paleodeclination and paleolatitude of the Markam area (stippled) from Eurasian APWP of *Besse and Courtillot* [1991] and observed declination and paleolatitude from the Cuowa (K_1c) and Mangkang (K_2mn) Formations. Uncertainties are indicated by dashed lines.

nificant. The picture is in accord with the crustal shortening and thickening model for the Tibetan Plateau proposed by *Dewey et al.* [1988]. This model regards the eastern zone as a giant right-lateral transpressional zone and the western zone as a giant left-lateral transpressional zone (see Figure 1). The displacement associated with Indian indentation within these transpressional zones was probably accommodated mainly by block rotation, which may lead to east-west, instead of north-south, compression in eastern Tibet [*England and Molnar*, 1990]. Crustal shortening in the region may be due to east-west horizontal shortening and vertical stretching, which cannot be detected by means of paleomagnetics. This mechanism appears to be compatible with the northeast-convex arcuate shape of the major faults and east-west compressional structural features in eastern Tibet [*Wang and Chu*, 1988].

CONCLUSION

Prefolding paleomagnetic directions have been isolated from the Lower and Middle Cretaceous redbed formations from eastern Qiangtang Terrane of Tibet. The ChRMs from these Cretaceous formations are most probably of Cretaceous age. Both formations reveal significant clockwise rotation but little latitudinal change. The existing paleomagnetic data from Tibet and its environs appears to be compatible with a crustal shortening/thickening model for the Tibetan Plateau [*Dewey et al.*, 1988] in that the continental crust right in front of the Indian indenter was substantially shortened in a north-south direction whereas tectonic rotation was predominant with insignificant latitudinal crustal shortening in the western and eastern zones bounding the indenter. However, more data is needed both from the plateau per se and from the western and eastern bounding zones to further discriminate between this model and the lateral eastward crustal extrusion model [*Molnar and Tapponnier*, 1975].

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